LUCAS MOUGEOT

AI Systems Engineer | High-Assurance LLM Infrastructure & Materials Science

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SUMMARY

Systems Engineer with an M.S. in Materials Science (NASA-funded) specializing in high-assurance AI infrastructure. Expert at architecting deterministic RAG pipelines, mitigating LLM hallucinations, and bridging complex physical sensor data with production-grade Python/AWS backends. Proven 'Founding Engineer' track record building scalable ingestion engines and distributed systems that remain reliable under real-world and regulated constraints.

EXPERIENCE

Founding Systems Engineer — Stealth Platform 2024 – Present

- Built and operate a production full-stack compliance platform (React, TypeScript, Python, PostgreSQL) supporting structured intake workflows, backend APIs, background processing, and rule-based evaluation services.
- Designed and implemented a deterministic Python codebase ingestion engine that compiles arbitrary repositories into immutable structural and dependency graphs using AST-based import analysis, enabling reproducible context compression for LLM-driven coding, classification, and decision workflows.
- Designed a deterministic ingestion engine for high-accuracy PDF/Image data extraction; implemented hybrid RAG pipelines (vector + full-text) with citation-anchored retrieval to eliminate hallucinations and ensure 100% auditability.
- Enforced schema-first workflow design to isolate probabilistic AI components behind deterministic validation layers, ensuring auditability, replayable state transitions, and dispute-grade traceability.
- Designed versioned legal knowledge graphs in PostgreSQL (ltree) with temporal validity windows and override chains to support historical reconstruction and regulatory auditability.
- Containerized multi-service architecture (Docker, PostgreSQL, Redis, BullMQ) and built regression, fuzz, and concurrency test suites to guarantee stable behavior under parallel workloads.

Integration Engineer — EZ Systems (Full-Time) Feb 2025 – Aug 2025

- Served as an R&D and systems integration engineer at the intersection of AI, robotics, AR/VR, and physical compliance constraints.
- Architected a unified QA and deployment framework integrating AI inference, sensor calibration, containerized services, and CI/CD pipelines, reducing pre-deployment validation time and ensuring reproducible system behavior.
- Developed robotic scan-path generation software from customer-provided STEP models, translating geometric requirements into executable motion plans.
- Led engineering for a self-driving mobile platform pilot operating in solar-field environments, addressing sensing reliability, deployment constraints, and safety validation.
- Authored technical strategy documents, including a federal grant proposal on explainable AI, internal R&D specifications, and company-wide AI integration guides.

Graduate Researcher — Materials Science (NASA-Funded Capillary Devices) 2023 – 2024

- Designed and fabricated porous ceramic devices enabling passive capillary transport in microgravity environments for space agriculture.
- Developed 36 ceramic formulations with controlled pore structure; executed iterative experiments to optimize capillary rise, stability, and manufacturability.
- Built MATLAB-based computer-vision and data analysis pipelines for pore segmentation, morphology quantification, and statistical ranking of candidate formulations.
- Characterized microstructure and composition using SEM/EDS; documented processing–structure–property relationships.

Undergraduate Researcher — Computational Geometry (NSF-Funded) 2020 – 2022

- Implemented bounded-degree planar graph constructions in modern C++ using CGAL under strict geometric constraints.
- Designed locality-preserving spatial algorithms for dense and heterogeneous point datasets.
- Contributed to peer-reviewed ACM Journal of Experimental Algorithmics publication.

SELECTED PROJECTS

Automated Lithium-Ion Battery Pack Assembler Senior Design Capstone

- Designed and implemented an automated lithium-ion battery pack assembly system using an AutomationDirect Do-More PLC.
- Developed ladder logic coordinating motion sequencing, interlocks, fault handling, and safe state transitions.
- Integrated an RP Series robotic dispenser via RS-232 serial communication to initiate external job file execution.
- Implemented PWM-based servo motor control for precise positioning and timing.
- Delivered a deterministic, repeatable automation pipeline with full I/O mapping and bring-up procedures prior to generative AI tools.

EDUCATION

University of North Florida

M.S. Materials Science & Engineering (Thesis-Based) — GPA: 4.00 / 4.00

- Thesis: Development and Characterization of Ceramic Devices for Capillary Action in Microgravity for the Purpose of Growing Vegetables in Space (NASA-Funded)
- Specialized in the characterization of physical sensing devices; deep experience with SEM/EDS and processing-structure-property relationships.

B.S. Electrical Engineering — GPA: 3.93 / 4.00

B.A. Interdisciplinary Studies: Morphology of Futurism — GPA: 4.00 / 4.00

Concentrations: Mechanical Engineering, Materials Science, Chemistry, Physics, Computing, Mathematics, Mandarin Chinese

PUBLICATIONS & PRESENTATIONS

- Bounded-Degree Plane Geometric Spanners in Practice — ACM Journal of Experimental Algorithmics (Peer-Reviewed)
- Highly Sensitive and Stable pH Sensor Electrodes of TiN Fabricated Using HiPIMS with Kick — AVS 69 Poster, 2023
- Interactive Tool for Bounded-Degree Plane Geometric Spanners — SoCG Media Exposition, 2021

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++

Data & AI: LLM Orchestration (LangChain/LlamaIndex), RAG, Context Engineering, Prompt Engineering, Hallucination Filtering, Structured Extraction (PDF/Vision), NLP, Vector Databases
Frontend: React, Next.js, TailwindCSS
Infrastructure: AWS (S3, Lambda, Bedrock), Docker, PostgreSQL, Redis, containerized service architectures, CI/CD pipelines, cloud deployments (AWS-based)
Data & Systems: API design, background workers, data pipelines, distributed systems, computational geometry
Robotics & Sensors: LiDAR integration, robotics motion planning, sensor calibration
Tools: GitHub Actions, Supabase, BullMQ, Playwright